TASK SPECIFICATION: P6-07 PHASE P6 — ACADEMIC CAMPUS SERVICES & RESOURCE PODS

Study Rooms UI & Digital Evacuation Timer

Git Branch: `feature/P6-07-rooms-ui` · Target Pull Request: feature/P6-07-rooms-ui

Core Objective & Technical Scope:
Room catalog cards, live occupancy indicators, pod booking dialog, and prominent real-time countdown evacuation timer.

Priority Level	Medium	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P6-07-rooms-ui`
Upstream Dependencies	P6-06, P3-05		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Build RoomCatalog displaying pod capacity, equipment tags, and availability status.
2. Build RoomBookingModal supporting team member entry and duration selection.
3. Build EvacuationTimer displaying live countdown with color-coded warning thresholds.
4. Implement extension request and early release actions.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `frontend/src/features/rooms/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Matches master design showcase.
- Countdown timer decrements accurately without memory leaks or drift.

VERIFICATION & TEST COVERAGE STRATEGY

- RoomCatalog component tests
- EvacuationTimer test
- Playwright room reservation flow